

Asim Ahmed

Mathematics–Computer Science Student | Machine Learning, Data Science & AI
San Francisco, CA | asimah0005@gmail.com
GitHub | LinkedIn | Portfolio

EDUCATION

University of California, San Diego

Expected Mar. 2027

B.S. Mathematics–Computer Science; Minor in Cognitive Science

Relevant Coursework: Advanced ML Methods, Supervised ML, Artificial Intelligence, Statistical NLP, Probability

EXPERIENCE

Support Vectors

Jun. – Aug. 2026

AI Agents & Fine-Tuning Cohort

Completed an applied program covering LLM agents, MCP tooling, DSPy optimization, and parameter-efficient fine-tuning; capstone: **The Disclosure Verifier** (below).

TECHNICAL SKILLS

Languages: Python, SQL

Machine Learning & Data: scikit-learn, PyTorch, CatBoost, pandas, NumPy, Conformal Prediction

Generative AI: DSPy, MCP, LangGraph, OpenAI SDK, Unsloth/QLoRA, GRPO/RLVR

Engineering: FastAPI, Streamlit, Git, GitHub Actions, pytest

PROJECTS

The Disclosure Verifier

Aug. 2026

Python, DSPy, MCP, Unsloth/QLoRA, GRPO (Hugging Face TRL), SEC EDGAR & XBRL, pytest, GitHub Actions

- Built an agentic SEC verification system with three MCP tool servers, a DSPy-optimized ChainOfThought extractor, budgeted checkpoint/resume orchestration, and QLoRA/GRPO (Hugging Face TRL) fine-tuning of Qwen2.5-7B-Instruct on a single RTX 4090.
- Backtested against 13,827 real SEC restatement fingerprints across 541 companies, matching 122 real prose instances and correctly flagging **118 of 122 (96.7%)** as inconsistent using only data that existed before each restatement happened, at a projected cost of \$11.31 to scan every S&P 500 company's full annual 10-K.
- Raised extraction F1 from a 0.730 manually designed prompt baseline to 0.784 via DSPy optimization, then cut the false-consistent rate from 0.148 to 0.036 (accuracy 0.766 → **0.855**) via RLVR/GRPO fine-tuning; validated the deterministic reconciler against a naive LLM-only judge, winning 15/15 adversarial cases with zero false positives, backed by 260 automated tests.

AI Sentinel

Sep. 2026

Python, FastAPI, OpenTelemetry, SQLite, pytest, systemd

- Built a reliability engine for AI services that detects failures against a rolling baseline, diagnoses the causing pipeline stage, and verifies each one-click fix on real traffic with automatic rollback and verified fixes replayed as regression tests; deployed persistently on a shared GPU cluster behind 138 automated tests.
- Designed a blind fault-injection eval that exposed a baseline-exclusion bug in the system's own detector (68% diagnosis accuracy at full scale), then re-ran it after each of four fixes to reach **98%** (49/50 trials; tuned on the same eval, not a held-out estimate).

NBA Win-Probability Prediction Platform

Aug. 2026

Python, pandas, scikit-learn, CatBoost, FastAPI, Streamlit, SQLite, pytest, Ruff, GitHub Actions

- Engineered an 11-season NBA win-probability pipeline with chronological splitting, a custom Elo engine, and a leakage-safe roster-continuity feature proxying player injuries from game logs; selected logistic regression over CatBoost, reaching **0.738 ROC-AUC** and 67.8% accuracy on a frozen 2,460-game test set.
- Deployed a FastAPI prediction service and Streamlit pick-the-winner dashboard using a shared feature-computation layer across training and inference, with a scheduled GitHub Actions pipeline keeping live features current; validated by 312 automated tests.
- Built a walk-forward backtest across 8 season cutoffs and a live calibration monitor scoring real user picks against the frozen model, surfacing a **1.7x calibration-error gap** between back-to-back and rested games hidden by the aggregate accuracy number.

Multi-Agent Vacation Planner

Aug. 2026

Python, LangGraph, Streamlit, OpenAI SDK, pytest, GitHub Actions

- Architected a multi-agent trip-planning system in LangGraph with parallel flight/activity search, a budget-dependent sequential hotel stage, and a final itinerary compiler, migrating from a mock-data prototype to **four live, keyless data sources** at zero API cost.
- Diagnosed and patched a parsing bug in a third-party scraping library that silently discarded valid results, and built an offline airport-disambiguation resolver with a hub-connection fallback for routes with no direct itinerary.
- Built a fact-checking guard that cross-references LLM-generated recommendations against ground-truth data to prevent hallucinated place names, and a TTL cache that cut repeat search latency **from 1.15s to 0.00s**, backed by 102 automated tests, CI via GitHub Actions, and a public Streamlit Community Cloud deployment.